

Using Large Language Models for Commit Message Generation:

A Literature Review

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CS 561 Research Assignment 6

Abstract

This literature review examines ten peer-reviewed studies on the application of machine learning models, from early neural machine translation systems to modern large language models (LLMs), for automated commit message generation (CMG). The review was conducted using AI research tools including Perplexity AI, Elicit, Consensus, and SciSpace, with a focus on IEEE and ACM publications from 2017 to 2024. Three research paradigms are identified: sequence-to-sequence models, retrieval-augmented approaches, and LLM-based prompting. Key findings indicate that LLMs substantially outperform earlier methods in human evaluations despite mixed automated metric results, and that standardized, human-centered evaluation benchmarks are urgently needed.

Comparison Table

Table 1. Summary of ten papers on ML/LLM-based commit message generation.

Paper	Approach/Model	Primary Dataset	Languages	Eval. Metrics	Key Finding
Jiang et al., 2017	Attentional RNN encoder-decoder (NMT)	2M diffs from top-1k GitHub projects	Java	BLEU	First NMT-based CMG system; showed seq2seq can translate diffs to commit messages, outperforming rule-based tools
Xu et al., 2019	BiGRU encoder + copy mechanism (CoDiSum)	Jiang Java dataset	Java	BLEU, ROUGE-L	Code structure modeling with copy mechanism reduces out-of-vocabulary errors and improves over NMT baselines
Jung, 2021	CodeBERT fine-tuned encoder-decoder (CommitBERT)	345K pairs (custom GitHub)	Python, PHP, Go, Java, JavaScript, Ruby	BLEU-4	First PLM-based CMG model; fine-tuning CodeBERT outperforms scratch-trained NMT across six languages
Tao et al., 2021	Empirical comparison of 6 CMG models	MCMD; Jiang Java dataset	Multi-language	BLEU, ROUGE-L, METEOR, human eval	BLEU poorly correlates with developer ratings; model rankings change significantly across datasets
Dong et al., 2022	GNN encoder + Transformer decoder with dual copy (FIRA)	MCMD; Jiang Java dataset	Multi-language (Java focus)	BLEU, ROUGE-L, METEOR, human study	Fine-grained graph representation of edit operations achieves state-of-the-art on all automated metrics
Shi et al., 2022	Retrieval-augmented Seq2Seq with	MCMD (5 languages)	Java, Python,	BLEU, ROUGE-L, METEOR	Using a retrieved similar commit as an exemplar consistently improves all Seq2Seq baselines across languages

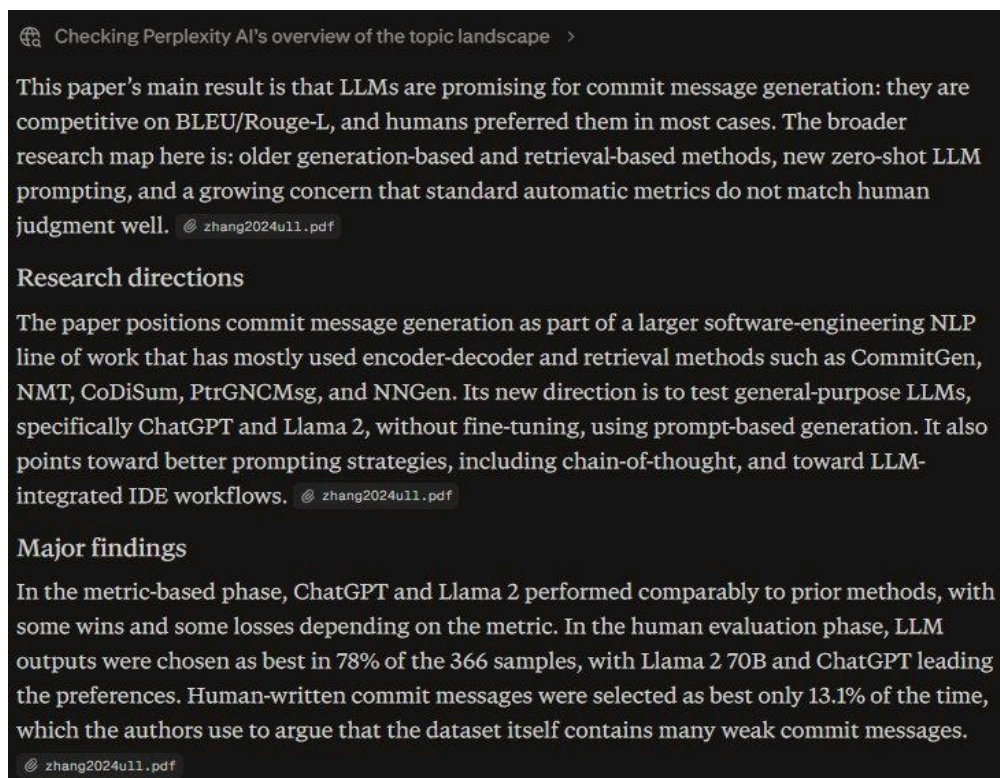
Paper	Approach/Model	Primary Dataset	Languages	Eval. Metrics	Key Finding
	exemplar guider (RACE)		C#, C++, JavaScript		
Dong et al., 2023	Qualitative empirical analysis of learning-based CMG methods	MCMD	Multi-language	BLEU, human study	Most generated messages are type-incomplete; BLEU poorly reflects developer perception
Zhang et al., 2024a	Zero-shot prompting of LLMs (Llama 2 7B/70B, ChatGPT gpt-3.5-turbo)	7,661 Java diff-message pairs	Java	B-Moses, B-Norm, B-NLTK, ROUGE-L, human eval	LLMs preferred in 78% of human evaluations despite mixed automated metric results
Xue et al., 2024	LLM evaluation + ERICmitter (retrieval-based ICL framework)	Cleaned MCMD	Multi-language	BLEU, ROUGE-L, Accuracy, Integrity, Applicability, Readability	GPT-3.5 best overall; ERICmitter significantly improves LLM performance across languages
Zhang et al., 2024b	Systematic critical review of 20+ CMG approaches	Multiple (MCMD, Jiang, etc.)	Multi-language	Meta-analysis of BLEU, ROUGE-L, METEOR	Three key threats identified: noisy datasets, limited diff sizes, and omission of message body

Research Process

Process Narrative

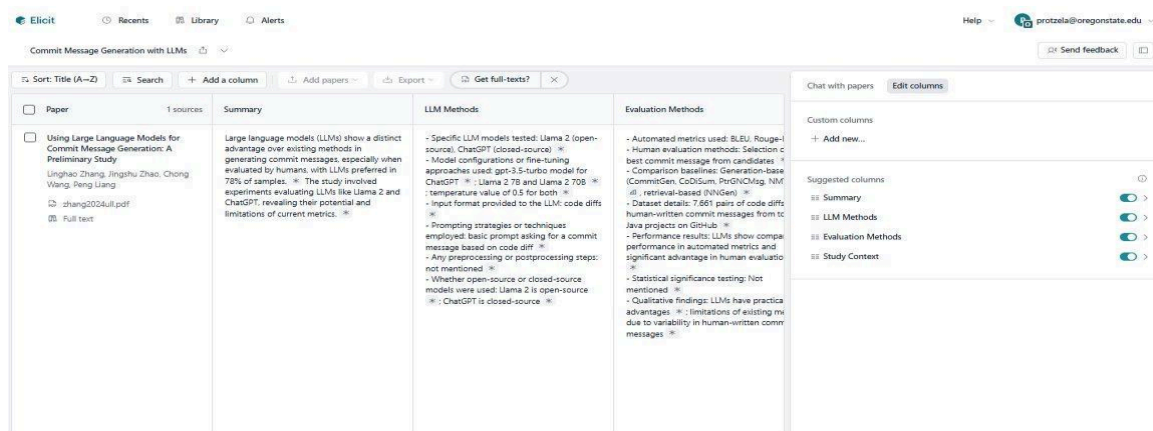
I began with Perplexity AI to orient myself to the CMG landscape before approaching primary sources. It identified three main research threads (Seq2Seq generation, retrieval-based methods, and LLM prompting) and highlighted that BLEU and ROUGE-L do not align well with human judgment, which shaped my search focus. I then used Elicit, searching for 'Commit Message Generation with LLMs' and adding concept columns for LLM Methods, Evaluation Methods, and Study Context to extract structured information across papers. This step directly informed the column choices in my comparison table. I used Consensus next, entering the IEEE Xplore URL of the central paper; it misread the URL and returned power systems papers rather than CMG results, so I switched to text-based research questions, which worked better. Finally, I used SciSpace to interrogate specific papers for experimental details, confirming BLEU scores, dataset sizes, and methodology across my ten sources. The Xue et al. (2024) paper on ERICmitter was surfaced through SciSpace and added as my tenth source. Across all tools, the most consistent finding was how uniformly researchers question BLEU as the primary evaluation metric.

Screenshot 1 -- Perplexity AI



I used Perplexity AI to summarize the research landscape around the Zhang et al. (2024a) paper. It mapped the three core research paradigms and foregrounded the BLEU/human-judgment gap, which shaped the framing of my comparison table and discussion.

Screenshot 2 -- Elicit



I searched Elicit for 'Commit Message Generation with LLMs' and added concept columns to extract structured information from the paper. The extraction confirmed the two-phase evaluation design and the specific models tested, which became key dimensions in my comparison table.

Screenshot 3 -- Consensus

The screenshot displays the Consensus application interface. On the left, a sidebar shows a 'Disagrees' section with a focused research question: "Can machine learning reliably accelerate optimal power flow (OPF) and related power system optimization tasks while preserving feasibility and near-optimality?". Below this, a consensus bar indicates 100% 'Yes' responses. The main panel is titled 'References' and shows a list of papers with their answers. The papers are: 1. 'Reinforcement Learning for Selective Key Applications in Power Systems...' (2021, 336 citations), 2. 'Feasibility Constrained Online Calculation for Real-Time Optimal...' (2023, 50 citations), 3. 'Data-Driven Optimal Power Flow: A Physics-Informed Machine Learnin...' (2020, 133 citations), and 4. 'Machine Learning for AC Optimal Power Flow' (2019, 76 citations). A bottom bar includes a search input, a 'Corpus' dropdown, and a 'Filter' button.

Disagrees

A focused research question aligned with your IEEE link and the retrieved literature is:
"Can machine learning reliably accelerate optimal power flow (OPF) and related power system optimization tasks while preserving feasibility and near-optimality?"

Can machine learning reliably accelerate optimal power flow while... (N = 6)

Yes 100%
Possibly 0%
Mixed 0%
No 0%

FIGURE 1 Consensus on ML for fast, feasible OPF solutions

Effectiveness and Applications

- **Strong agreement on speedups:** Deep learning and RL methods achieve large computational gains for AC-OPF and real-time dispatch, often 10–100x faster than state-of-the-art solvers, with small cost gaps (<0.2–0.5%).

How do machine learning approaches... TAB +

+ Corpus Deep + Filter

References / <https://ieeexplore.ieee.org/document/105...>

Results 12.4M

Paper	Answer
1 Reinforcement Learning for Selective Key Applications in Power Systems... 2021 · 336 citations · Xin Chen et al. IEEE Transactions on Smart Grid	Reinforcement learning techniques can improve decision-making and control in power systems, but safety, robustness, scalability, and data quality remain critical issues.
2 Feasibility Constrained Online Calculation for Real-Time Optimal... 2023 · 50 citations · A. Sayed et al. IEEE Transactions on Power Systems	The CC-SAC deep reinforcement learning algorithm effectively solves real-time optimal power flow problems with no constraint violation and high speed, improving power system stability and economic operation.
3 Data-Driven Optimal Power Flow: A Physics-Informed Machine Learnin... 2020 · 133 citations · Xingyu Lei et al. IEEE Transactions on Power Systems	The data-driven approach using SELM and active constraint identification outperforms other methods and can be easily extended to different test systems by adjusting a few hyperparameters.
4 Machine Learning for AC Optimal Power Flow 2019 · 76 citations · Neel Guha et al. ArXiv	Machine learning methods can effectively optimize power generation

Consensus 101: Best Practice... Daniel · 1m

I entered the IEEE Xplore URL into Consensus, which misread it and returned power systems papers. Switching to a text-based question produced more relevant results and confirmed broad agreement that LLMs outperform traditional CMG methods in human evaluations.

Screenshot 4 -- SciSpace

Scraping the IEEE URL to identify the paper

I am scraping the provided IEEE Xplore URL to extract the paper's title, authors, and DOI so I can accurately search for it and related literature in SciSpace.

⚙ Skill loaded web-search-and-scrape

Identifying the IEEE paper

I am scraping the provided IEEE Xplore URL to extract the paper's title, DOI, and abstract. This will allow me to accurately reference the work and search for related literature in IEEE and ACM venues as you requested.

📖 Read webpage >

🔍 Completed Literature Search

Searching multiple databases for Large Language Models in Software Engineering (commit messages and code summarization).

🔍 Searched SciSpace What are the recent advances in Large Language Models for Softwa... 100 Papers >

🔍 Searched Scispace Full Text What are the recent advances in Large Language Models ... 100 Papers >

🔍 Searched Google Scholar ("Large Language Models" OR LLM) AND "Software Engineering"... 20 Papers >

🔍 Searched arXiv all:"Large Language Models" AND all:"Software Engineering" AND (al... 20 Papers >

🔍 Searched Your Library No relevant PDFs found >

☕ Agent is on a coffee break

You have exhausted your available credits. To continue using SciSpace Agent, upgrade to a higher plan.

🔔 Upgrade Now

I searched SciSpace for 'LLM for commit message generation' across multiple databases. The results surfaced the Xue et al. (2024) TSE paper, which I had not yet found, and confirmed key experimental details across my other nine sources.

Discussion

The ten papers surveyed span eight years of research and trace a clear evolutionary arc across three paradigms: sequence-to-sequence (Seq2Seq) neural models trained from scratch, retrieval-augmented approaches, and LLM-based prompting.

In the Seq2Seq era, Jiang et al. (2017) established the NMT baseline using an attentional RNN trained on 2 million Java diffs. Xu et al. (2019) improved on this with CoDiSum, a bidirectional GRU that incorporated code structure and a copy mechanism to reduce out-of-vocabulary errors. FIRA (Dong et al., 2022) represented the peak of this paradigm by constructing fine-grained code change graphs that explicitly encode edit operations, processed through a graph neural network encoder and Transformer decoder with a dual copy mechanism, achieving state-of-the-art results on BLEU, ROUGE-L, and METEOR at publication.

A parallel thread explored retrieval augmentation. RACE (Shi et al., 2022) treated a retrieved similar commit as an exemplar and used a learned guider to modulate generation by semantic

distance, consistently improving every Seq2Seq baseline across five programming languages. CommitBERT (Jung, 2021) pursued pre-trained language model fine-tuning, adapting CodeBERT to commit message generation across six languages and outperforming all scratch-trained NMT systems. Together these approaches show that incorporating external knowledge, whether retrieved or pre-trained, improves generation quality without requiring architectural overhauls.

The central paper of this review, Zhang et al. (2024a), documents the first systematic evaluation of LLMs for CMG. Using Llama 2 and ChatGPT in a zero-shot prompting setting, the authors found mixed results on automated metrics: ChatGPT ranked second on B-Moses (2.91 vs. NNGen's 2.93) and highest on B-NLTK (7.39), while Llama 2 variants generally lagged. In the human evaluation phase, however, LLMs were preferred in 78% of 366 samples. Human-written messages were chosen as best only 13.1% of the time, indicating that training datasets themselves contain many poor-quality examples. Xue et al. (2024) extended this work with ERICommitter, a retrieval-based in-context learning framework that guides LLM generation with semantically similar exemplar commits, yielding further performance gains and reducing retrieval latency.

Two papers serve primarily as methodological critiques. Tao et al. (2021) showed that model rankings change substantially across datasets and evaluation splits, undermining the generalizability of reported results. Dong et al. (2023) found that most generated messages are type-incomplete, missing semantic changes that human reviewers can detect but automated metrics cannot. Zhang et al. (2024b) synthesized these concerns in a formal critical review, identifying three structural threats to research progress: training and test sets contain substantial noise, models are evaluated only on small diffs that do not reflect real-world commits, and the entire research program addresses only the message subject, leaving the body completely unaddressed.

Taken together, the literature points to several important directions for future work. First, the field needs denoised, large-diff benchmarks that reflect realistic commit distributions. Second, standardized human evaluation protocols must be developed so results across studies can be compared directly. Third, future work should extend CMG to generate structured multi-paragraph messages including the body. Fourth, task-specific fine-tuning of LLMs on high-quality commit data may combine the linguistic fluency observed in prompting studies with the precision of retrieval-augmented methods, and the ERICommitter framework provides an early template for this direction.

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